

DS-03 — Product Affinity, Cross-Sell & Basket Analytics

Dataset ID: DS-03

Purpose: Quantify product relationships so strategies can test cross-sell, bundles, basket growth and product combinations.

Instruction alignment: Supports frequency growth, basket growth, product mix optimisation, cross-sell and product-specific marketing. Relationships are hypotheses supported by observed purchase combinations.

Provenance & modelling note

This is a synthetic quantitative extension created for the Vayal Foods simulation. It is designed to be internally consistent with the supplied Vayal company documents and to fill quantitative evidence gaps identified by the strategy/instruction documents. It must not be treated as an audited source document. Known company totals are preserved where stated; newly introduced granular fields are modelling data.

Product Pair / Affinity Matrix

Pair penetration is measured against 44,359 customers with recognised purchase activity. Attach rate is the share of customers buying the complementary product among customers buying the primary product.

Primary Product	Complementary Product	Customers Buying Both	Pair Penetration of Purchased Customers	Attach Rate Among Primary Buyers
P1	P2	8,428	0.19	0.34
P1	P3	7,097	0.16	0.29
P1	P4	6,210	0.14	0.25
P1	P5	3,105	0.07	0.13
P2	P4	4,879	0.11	0.22
P2	P3	3,549	0.08	0.15
P2	P5	2,218	0.05	0.09
P3	P4	3,992	0.09	0.19
P3	P5	1,774	0.04	0.08
P4	P5	3,549	0.08	0.16
P5	P1	4,436	0.1	0.18
P5	P2	2,662	0.06	0.12
P4	P1	5,767	0.13	0.24
P3	P1	7,097	0.16	0.29
P2	P1	8,428	0.19	0.34